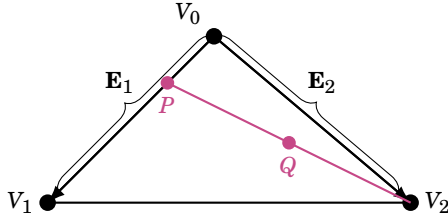


1 Triangle Parameterization via Barycentric Coordinates



Let a 3D triangle have vertices $V_0, V_1, V_2 \in \mathbb{R}^3$. Then any point along edge V_0V_1 can be written via linear interpolation as:

$$\begin{aligned} P &= (1-u)V_0 + uV_1, & Q &= (1-v)P + vV_2 \\ & & &= \underbrace{(1-v)(1-u)}_{\alpha} V_0 + \underbrace{(1-v)u}_{\beta} V_1 + \underbrace{v}_{\gamma} V_2 \end{aligned}$$

Observe that $\alpha + \beta + \gamma = (1-v)(1-u) + (1-v)u + v = 1$. Therefore any point Q with:

$$\boxed{Q = \alpha V_0 + \beta V_1 + \gamma V_2, \quad \alpha + \beta + \gamma = 1, \quad 0 \leq \alpha, \beta, \gamma \leq 1} \quad (1.1)$$

lies inside the triangle. With some algebra, this definition can be written in terms of the edges $E_1 = V_1 - V_0$, $E_2 = V_2 - V_0$.

$$\begin{aligned} Q &= (1-v)\left((1-u)V_0 + uV_1\right) + vV_2 \\ &= (1-v)\left(V_0 + uE_1\right) + vE_2 \\ &= V_0 + uE_1 - vV_0 - uvE_1 + vV_2 \\ &= V_0 + \underbrace{(u-uv)}_{v'} E_1 + \underbrace{v}_{v'} E_2 \end{aligned}$$

Once again, in order for Q to lie inside the triangle, we formulate two constraints:

$$\begin{aligned} u' = u(1-v) \geq 0, \quad v' = v \geq 0, \quad u' + v' = v + v - uv \\ \leq 2 - 1 = 1 \end{aligned}$$

0 and 1 are extrema. Each equality occurs when:

$$\begin{aligned} u' = 0 &\Rightarrow u = 0 \vee v = 1, \quad \vee \quad u' + v' = 1 \Rightarrow \\ &(u = v = 0 \Rightarrow Q \equiv V_0) \vee \\ &(u = v = 1 \Rightarrow Q \equiv V_2) \vee \\ &(u = 1 \wedge v = 0 \Rightarrow Q \equiv V_1) \vee \\ &(u = 0 \wedge v = 1 \Rightarrow Q \equiv V_2) \end{aligned}$$

Therefore, the extrema occur when the ray hits the triangle at its vertices. To summarize, any point Q that lies in triangle $\triangle V_0V_1V_2$ with edges $E_1 = V_1 - V_0$, $E_2 = V_2 - V_0$ can be written as:

$$\boxed{Q = V_0 + u'E_1 + v'E_2, \quad u', v' \geq 0, \quad u' + v' \leq 1}$$

2 Ray-Triangle Intersection

A ray is given by

$$R(t) = O + tD, \quad t \geq 0,$$

where O is the origin and D is the direction (unit) vector. The parameter t describes the signed distance of how far along the ray a point lies. To find an intersection between the ray and the triangle, we equate the two parameterizations:

$$O + tD = V_0 + uE_1 + vE_2 \Rightarrow$$

$$\underbrace{O - V_0}_S = uE_1 + vE_2 - tD \Rightarrow$$

$$\underbrace{\begin{bmatrix} E_1 & E_2 & -D \end{bmatrix}}_M \begin{bmatrix} u \\ v \\ t \end{bmatrix} = S$$

The latter expressions the intersection as a 3×3 linear system for the unknowns (u, v, t) .

3 Solving the System Using Cramer's Rule and Triple Products

In general, the determinant of $[A \mid B \mid C]$ is defined as $\det([A \mid B \mid C]) = A \cdot (B \times C)$. The determinant of M is therefore scalar triple product

$$a = \det(M) = E_1 \cdot (E_2 \times (-D)) = E_1 \cdot (D \times E_2)$$

3.1 Determinant and Parallelism Test

If $a \approx 0$, then $\text{rank}(M) < 3$, i.e. some of E_1, E_2, D are linearly dependent, which occurs when

- the ray is parallel to the triangle plane.
- two of the triangle sides are parallel, meaning the triangle is degenerate.

$$\therefore \boxed{a = E_1 \cdot (D \times E_2) \approx 0 \Rightarrow \text{no stable intersection}}$$

From Cramer's rule, we can now solve for u, v, t using the determinant formulas.

3.2 Cramer's Rule for u, v, t

From Cramer's rule:

$$u = \frac{\det(\begin{bmatrix} S & E_2 & -D \end{bmatrix})}{\det(M)} = \frac{S \cdot (D \times E_2)}{a} \stackrel{A \cdot (B \times C) = B \cdot (C \times A)}{=} \frac{D \cdot (E_2 \times S)}{a}$$

We often define $h \equiv E_2 \times S$ so u is given by:

$$\boxed{u = \frac{D \cdot q}{a}, \quad h = E_2 \times S, \quad a = E_1 \cdot (D \times E_2)}$$

Again, by Cramer's rule and using the triple product properties we can arrive at:

$$\boxed{v = \frac{\det(\begin{bmatrix} E_1 & S & -D \end{bmatrix})}{a} = \frac{D \cdot q}{a}, \quad q = S \times E_1}$$

t is the last parameter and it's computed the same way:

$$\boxed{t = \frac{\det(\begin{bmatrix} E_1 & E_2 & S \end{bmatrix})}{a} = \frac{E_2 \cdot q}{a}, \quad q = S \times E_1}$$

3.3 Cramer's Rule for v

Replace the second column with S :

$$v = \frac{\det([E_1 \ S \ -D])}{a}.$$

Using the scalar triple product identity

$$A \cdot (B \times C) = B \cdot (C \times A) = C \cdot (A \times B),$$

we rewrite

$$\det([E_1 \ S \ -D]) = D \cdot (S \times E_1).$$

Define the helper vector

$$q = S \times E_1,$$

so that

$$v = \frac{D \cdot q}{a}.$$

3.4 Cramer's Rule for t

Replace the third column with S :

$$t = \frac{\det([E_1 \ E_2 \ S])}{a}.$$

Using cyclic permutation of the triple product,

$$t = \frac{E_2 \cdot (S \times E_1)}{a} = \frac{E_2 \cdot q}{a}.$$

4 Geometric Interpretation of h, a, t

4.1 The Vector $h = D \times E_2$

The vector h is perpendicular to the plane spanned by the ray direction and the triangle edge E_2 .

4.2 The Scalar $a = E_1 \cdot h$

The value a is the signed volume of the parallelepiped formed by (E_1, D, E_2) . Its magnitude is large when D is perpendicular to the triangle plane (spanned by E_1, E_2 and small when the ray is nearly parallel to the triangle plane. Its sign encodes the orientation relative to the triangle plane (useful for backface culling).

u and v are normalized measure how far along we need to span along edges E_1 and E_2 in order to find the hit point as a linear combination $V_0 + uE_1 + vE_2$.

4.3 Meaning of t

The parameter t expresses the distance along the ray direction D from the ray origin O to the intersection point:

$$P = O + tD.$$

If $t < 0$, the intersection lies “behind” the ray origin and is rejected.

5 Pseudocode of the Möller–Trumbore Algorithm

Combining all these, we're ready to formulate a common algorithm for ray-triangle intersection.

Algorithm 1 Ray–Triangle Intersection (Möller–Trumbore)

Require: Ray origin O , direction D ; triangle vertices V_0, V_1, V_2

Ensure: Whether the ray intersects the triangle and the point of intersection

```
1:  $E_1 \leftarrow V_1 - V_0$ 
2:  $E_2 \leftarrow V_2 - V_0$ 
3:  $h \leftarrow D \times E_2$ 
4:  $a \leftarrow E_1 \cdot h$ 
5: if  $|a| < \varepsilon$  then
6:   return (false,  $\emptyset$ ) ▷ Ray parallel to triangle
7:  $S \leftarrow O - V_0$ 
8:  $u \leftarrow \frac{S \cdot h}{a}$ 
9: if  $u < 0$  or  $u > 1$  then
10:  return (false,  $\emptyset$ )
11:  $q \leftarrow S \times E_1$ 
12:  $v \leftarrow \frac{D \cdot q}{a}$ 
13: if  $v < 0$  or  $u + v > 1$  then
14:  return (false,  $\emptyset$ )
15:  $t \leftarrow \frac{E_2 \cdot q}{a}$ 
16: if  $t \leq \varepsilon$  then
17:  return (false,  $\emptyset$ ) ▷ Intersection behind ray origin
18:  $P \leftarrow O + tD$ 
19: return (true,  $P$ )
```
